

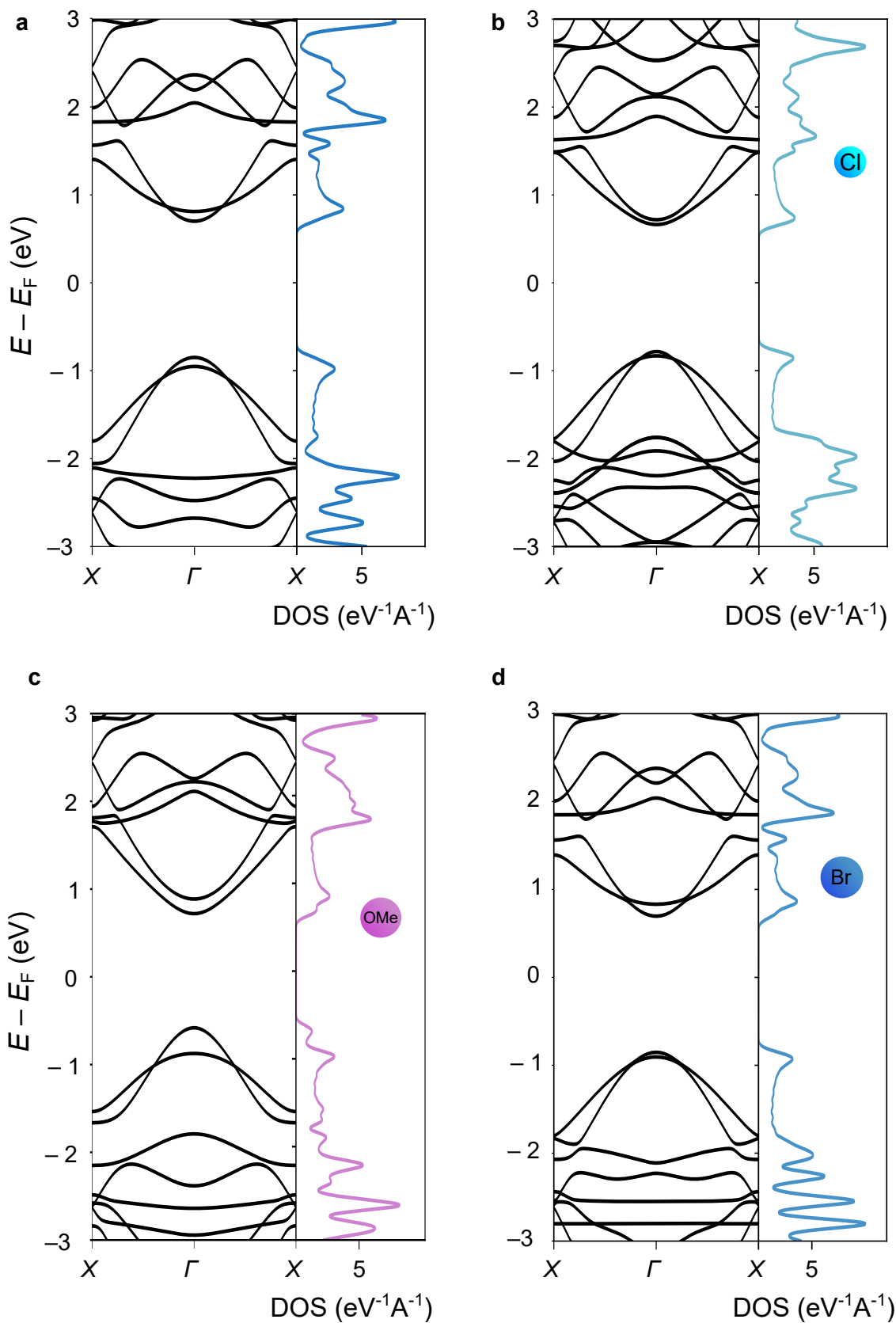


Figure 7.3: Band structure and density of states for graphene nanoribbons decorated with various electron donors or acceptors resulted after applying the GGA approximation for the exchange–correlation energy term.

resulting in a non-negligible interaction between the continuum of states in GNR and the localised state on the edge-attached adduct.

Furthermore, at this level of theory it appears that when anthracene groups are introduced along the edges the particle-hole symmetry is broken, as the band structure spectrum is no longer symmetric with respect to the Fermi energy.

Fig. 7.3 compares the band structures and DOS between GNR-H (top-left) and other chemical dopants. In the case of GNR-OMe, four electron-donating methoxy groups per repeating unit have been considered, while six negative inductive Chlorine groups were considered for GNR-Cl. Firstly, it is possible to observe the frontier bands in case of GNR-OMe are $\simeq 0.5$ eV higher than those of GNR-H, likely due to charge redistribution induced by the methoxy groups.

In addition to this, the DFT simulations suggest that the aromatic structure of GNR-OMe might be slightly distorted because of the steric repulsion introduced by the methoxy groups. In order to test this hypothesis, we performed a DFT simulation of a non-planar GNR-H having the same spatial distortion of GNR-OMe without the methoxy groups. The methoxy groups were replaced by hydrogen atoms after the geometric optimization of GNR-OMe, while keeping the distorted aromatic core fixed. The results suggest that the geometric distortions are responsible for only one-half of the bandgap shrinkage, i.e. 125 meV, while the other half can be ascribed to the larger effective width of the π -conjugated system of GNR-OMe due to the delocalization of the frontier states over the oxygen of the methoxy groups.

Interestingly, despite the fact that both the bromine and chlorine substitutions share electrons with the ribbon preferentially via σ -bond, GNR-Cl exhibits a more pronounced bandgap contraction (Tab. 7.1) and a decrease in effective mass m^* at the Γ point of the conduction band. The same holds true for the case of methoxy substitution. As the effective mobility μ_{eff} of charge carriers is proportional to the inverse of their effective mass, a decrease in the effective mass of the charge carriers may be advantageous for their transport, possibly resulting in better device performance. On the other hand, no appreciable deviation from the base case is observed for the ribbon functionalized with Br groups, which could be explained

Table 7.1: Calculated bandgaps using different exchange–correlation implementations.

<i>Edge functionalisation</i>	LDA – CA (eV)	GGA – PBE (eV)	ΔE_g (CA) (%)	ΔE_g (PBE) (%)
–H	1.57	1.55	–	–
–Anthracene	1.55	1.54	–0.96	–0.77
–Br	–	1.55	–	–0.04
–Cl	1.45	1.44	–7.62	–6.77
–OMe	1.31	1.3	–16.21	–16.1
–CH3	1.56	1.55	–0.44	–0.25
–CN	1.58	1.56	+0.62	+0.74
–NH2	1.36	1.35	–13.03	–12.9
–COOH	1.55	1.53	–0.84	–0.98
–NO2	1.47	1.47	–6	–5.24
–OH	1.47	1.46	–5.82	–5.82

by a lower electronegativity and consequently a lower ability to induce charge rearrangement of the ribbon’s frontier orbital.

7.3 Electrical Transport on Methox-Edge Decorated MGNRs

In order to test the idea of chemically-induced doping, we fabricated (see Ch. 3) two kinds of FETs using GNR–H (base case) and GNR–OMe (methoxy). As reported in Tab. 7.1, methoxy groups are predicted to induce the largest relative change in the bandgap compared to the base case, and the

GNR–OMe were provided by Dr Alicia Götz prof. Akimitsu Narita. From personal communications with them, the photoconductivity of GNR–OMe should be about 25% compared to the base case and a decreased effective mass.

We measured the transfer characteristics of hundreds of devices in ambient conditions and compared the base case with GNR–OMe. Overall the transfer characteristics do not observe a shift in terms of performance and the overall performance are poor.

Interestingly, chemical doping seems to induce a shift in the charge-neutrality point and the off state from negative to positive gate voltages. This might be due to the injection of holes from the methoxy into the nanoribbon, chemical doping often

also induced impurities in graphene in concentrations [14, 248]. A quantification of the concentrations of chemically-induced charge carriers is, however, impossible to estimate as a precise control of the number of functionalisations along the edges has not been achieved yet.

The $I_{\text{on}}/I_{\text{off}}$ ratio is at least three orders of magnitude from the requirements for digital electronics [23], and ON state delivers not enough density current for most of digital applications (Fig. 7.4).

Looking at the subthreshold slope figures at room temperature, the performance remains poor even at medium ($V_{\text{SD}} = 0.5 \text{ V}$) drain voltages, which indicates a poor control of the gate over the conduction channel. This might also be caused by the increased concentration of holes injected through the methoxy groups, an hypothesis which could be considered valid if we look at the overall worse performance as the drain voltage is increased. In fact, in the subthreshold region the drain current is proportional to the number of electrons over the source-channel barrier [249].

7.4 Substitutional Doping Through Heteroatoms Embedding in Graphene Nanoribbons Lattice

The idea of using chemical functionalisations to induce a preferential doping in carbon nanostructures is not new. Several proposals have indeed demonstrated the possibility to tune the conductance of ropes [46] and individual [47, 250] carbon nanotubes by introducing dopants in the walled structure, thus leading to *pn* junctions and Esaki diodes. These procedures, however, tend to be uncontrolled leaving the substrate with unknown number of dopants per carriers.

In contrast, bottom-up chemical synthesis may provide a viable route to structurally well-defined graphene ribbons with uniform structures in which the dopants are introduced along the edges through control of the chemical functionalisations. Amongst the various ribbon edges, cove GNRs have attracted a lot of interest for several reasons. First of all, their synthesis can be completely achieved in solution without any surface-assisted methodology. Secondly, the steric tension induced by the lateral groups on the structure makes them typically non-planar,

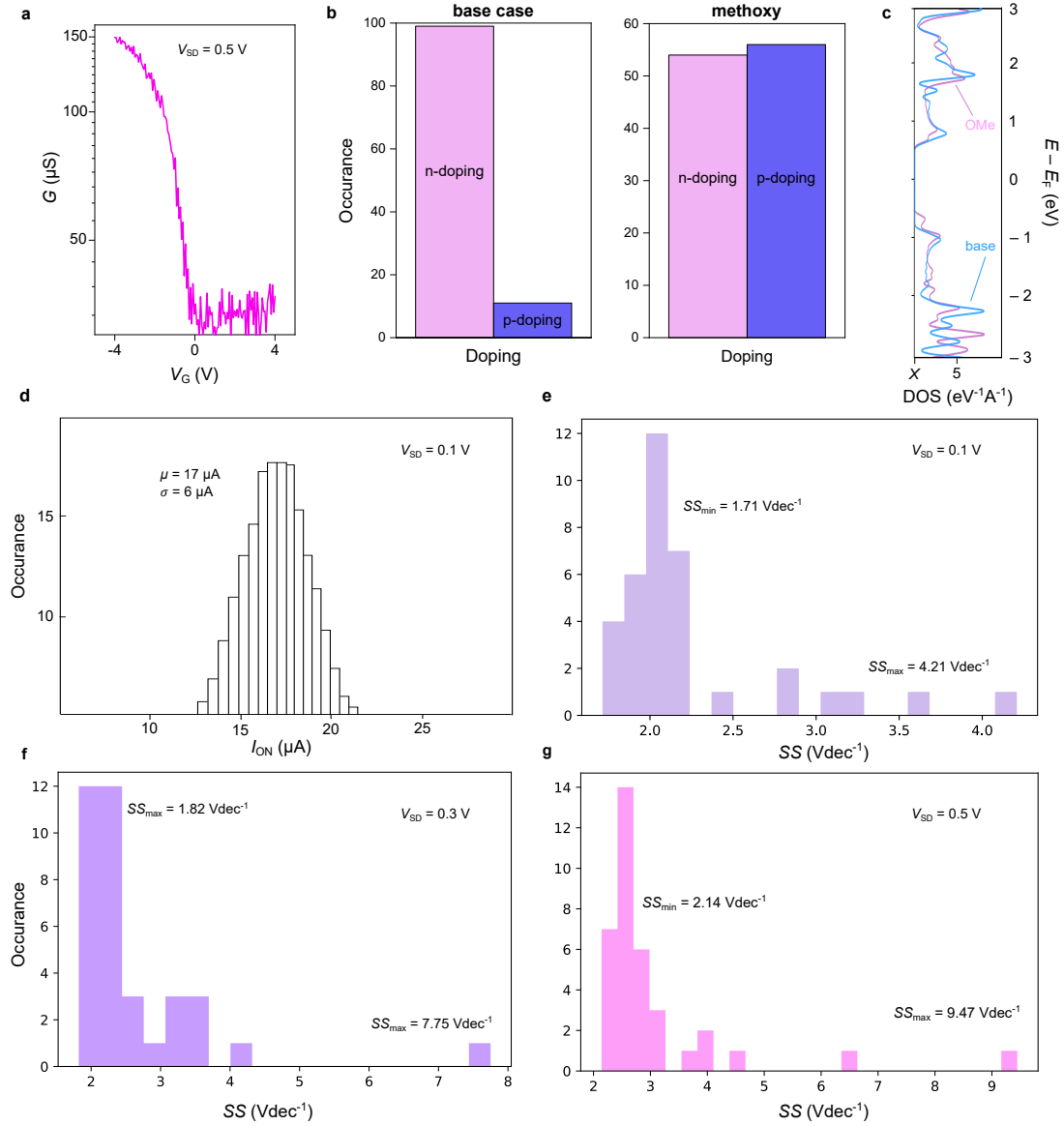


Figure 7.4: **a**, Typical transfer characteristic for GNR-OMe acquired at $V_{SD} = 0.5$ V. This particular device exhibits p-doping, and a $I_{on}/I_{off} \simeq 10^1$ **b**, Chemically doping with methoxy groups adds holes but also might add impurities compared to the base case, and there is a stastically relevant shift of the charge neutrality point from negative (n-doping) to positive (p-doping) gate voltages. **c**, When methoxy groups are introduced, the MGNR DOS experiences a bandgap narrowing of $\simeq -16.1\%$ **d**, Histogram of the output on-current (not normalised for the channel width) acquired at low drain voltages. **e**, Switching behaviour histograms at low and medium (**f,g**) biases.

open up possibility to investigate not only optical properties but also spin–orbit coupling effects in carbon nanostructures, which have remained so far elusive from experimental evidence.

Furthermore, theoretical calculations have shown that the wave functions for the frontier crystal orbitals are fully delocalized across the ribbon width, in contrast to zigzag-GNRs (with localized states at the ribbon edges), with electronic band gap of cove-edged GNR infinite ribbon settling around 1.70 eV, and mobilities values in the order of thousands $\text{cm V}^{-2} \text{s}^{-1}$ for both electrons and holes [92].

Feedback-controlled electroburning is used to produce nanogaps, localised at the bowtie constriction, with widths between 1–10 nm. $I_{\text{SD}} - V_{\text{SD}}$ characteristics are recorded for all devices immediately after nanogap formation and after drop-casting approximately 8 μL suspension of the pGNR hybrids.

Suspensions were prepared by sonicating a dry powder of the pGNRs in toluene (0.02 mg mL^{-1}) for 60 min. Several cycles of the electroburning protocol were performed, up to three different gate voltages for each device, and each time scanning in gate voltage at fixed bias to check for residual graphene quantum dots.

If any are found, the electroburning process is restarted at that gate voltage to clear the residual graphene. For all devices, the resistance limit was set at $2.3 \text{ G}\Omega$. Devices which show an increase in current post-deposition are selected for wire bonding and further measurements. Transport measurements were performed in an Oxford Instruments Triton 200 dilution refrigerator using low noise DC electronics. All measurements were performed at a temperature of 20 mK. Differential conductance is retrieved by numerical differentiation of the measured current.

The full charge-stability diagram of a typical device cooled down to 25 mK is shown in Fig. 7.5d. Yellow colours indicate regions of higher differential conductance while darker colours indicate lower regions of differential conductance. At gate voltages below -1 V clear signs of Coulomb blockade diamonds are present. Outside this gate voltage range, the level of current is close to the noise limit of the current-to-voltage converter used in our transport setup. In the absence of a discernible repeating pattern, it is possible that the faint diamonds visible at

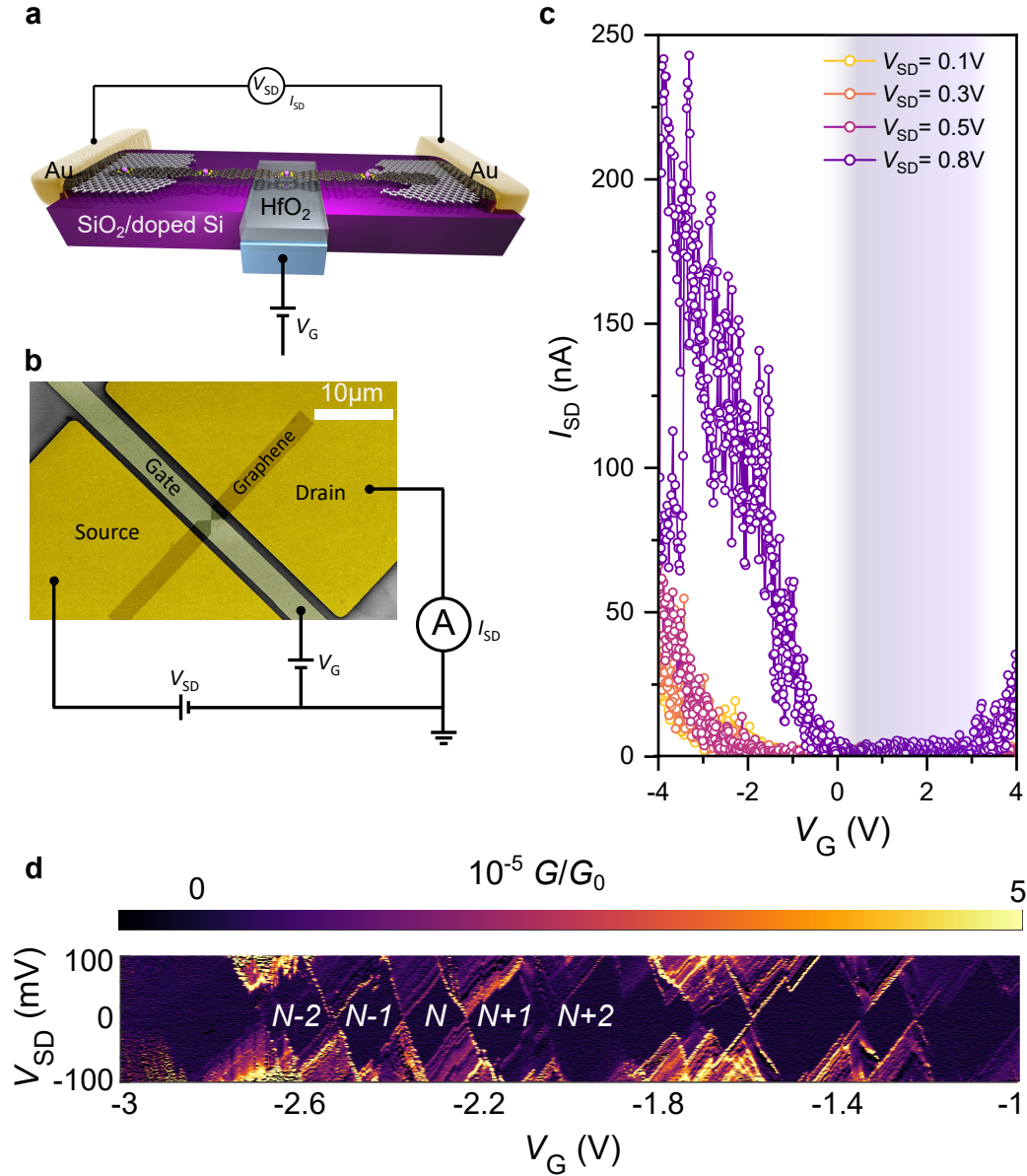


Figure 7.5: **a**, Cartoon of the device schematic and the electrical wiring. **b**, False-scale SEM image of a device sample, highlighting the Au pads, HfO₂ gate, graphene leads, and the electrical connections for biasing the device. **c**, Room-temperature FET response of a device under different biases. The device presents a bandgap in the positive region of V_G , demonstrating p-doping effect. The estimated $E_g \simeq 2\text{ eV}$ is in line with what we expect from optical absorption measurements. **d**, Clear demonstration of SET behaviour for porphyrin-GNR (pGNR) when these objects are cooled down at cryogenic ($T = 20\text{ mK}$) temperatures. The possibility of discriminating n-doping from p-doping at RT makes the mapping of the various charge states more easy.

Table 7.2: Lorentzian fit parameters extracted by fitting a low-bias gate trace. The two peaks are separated by 600 meV in voltage, which could be assigned to the single-electron charge transition $N \leftrightarrow N + 1$, where N is the number of electrons on the ribbons which is unknown. Reported errors are fitting errors.

Peak Position (eV)	Lifetime Broadening (meV)	$G_{\max}$ (G_0)
-1.64	2.5 ± 0.2	$(4.5 \pm 0.2) \times 10^{-7}$
-1.04	2.8 ± 0.2	$(2.8 \pm 0.1) \times 10^{-7}$

higher gate voltages are caused by spurious charges injected possibly through the gate oxide or surface states.

From Fig. 7.6 we can read the addition energies for the different charge states by reading the height of each diamond, within the assumption that only one electron is exchanged for each charge transition. Across all the diamonds, we can extract the gate coupling which measures the effectiveness of the gate voltage to change the chemical potential of the ribbon. We found $\alpha_G \simeq 0.65 \text{ eV V}^{-1}$ averaged across all the measurable charge states, with a slight asymmetric coupling with the drain lead for this particular device.

Furthermore, two peaks at $V_G \simeq -2.5 \text{ V}$ and $V_G \simeq -1.5 \text{ V}$ are closed, so we can extract their lifetime broadening by fitting a low-bias ($V_{SD} \simeq 2.6 \text{ mV}$) gate trace with a Breit–Wigner resonance (Tab. 7.2). The tunneling couplings with the graphene leads are one order of magnitude stronger than the majority of tunneling coupling reported for devices fabricated with normal graphene nanoribbons.

A possible explanation for the observed values for the tunneling coupling might be the different length of the pGNR compared to the bottom-up ribbons studied in the last two chapters. Since the thickness of the HfO_2 oxide on top of the gate is approximately 10 nm, we had to experimentally measure the capacitance of the the oxide in order to extract a more accurate value.

A parameter needed to determine the length of pGNRs is the relative permittivity of the gate oxide. Because of the ultra-thin nature of the oxide used in these devices, it is not possible to rely on the bulk value for HfO_2 and it must be experimentally determined [251]. This was achieved by microfabricating capacitors on the devices

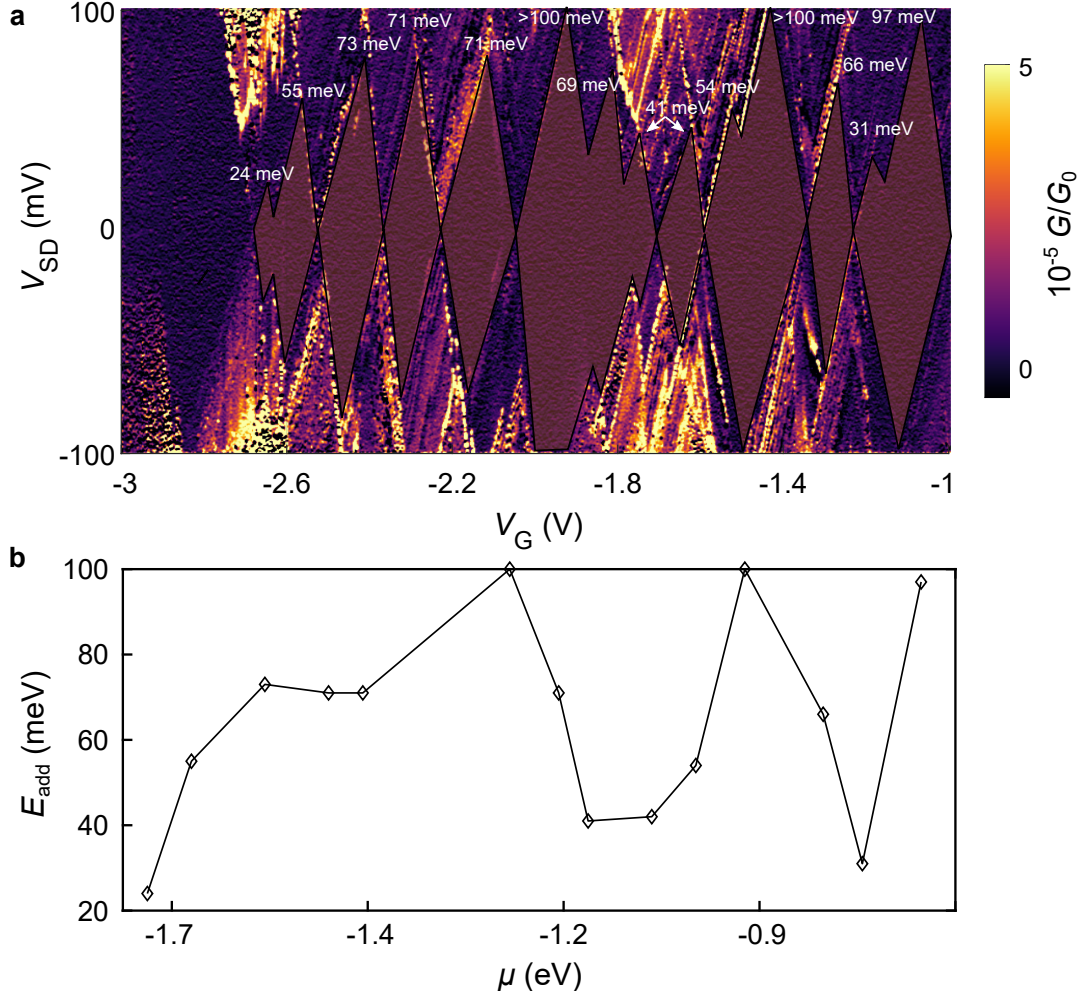


Figure 7.6: **a**, Low-temperature stability map demonstrating Coulomb blockade and well-defined charge states. **b**, We extract the addition energies from the diamond height and plot them as a function of the chemical potential felt by the ribbon as a result of the application of a gate voltage, assuming a constant gate coupling of $\alpha_G \simeq 0.65 \text{ eV V}^{-1}$ for all charge states.

using the following approach. A chip is selected from the same wafer used to fabricate devices 1, 2 and 3. The substrate is O_2 -plasma etched to remove the CVD graphene layer and prepare the surface for lithography (20 minutes, 25 sccm and 100 percent power). An array of top electrodes is patterned using electron-beam lithography, metal deposition and lift-off. The design consists of overlapping top electrode of varying areas with the gate contact pad separated by the ALD-deposited HfO_2 dielectric layer as shown in Figure 7. A small region is removed

Table 7.3: Measured values of electrode area, capacitance and ϵ_r for several capacitors. These are consistent across various randomly selected points onto a single chip. Errors for ϵ_r were found to be at the third decimal place.

Device name	Electrode Area (μm^2)	Measured Capacitance (pF)	ϵ_r
A1	7455	101.1 ± 1.0	15.3 ± 0.02
A2	4542	62.4 ± 0.6	15.51 ± 0.02
C3	2402	32.5 ± 0.3	15.26 ± 0.02
E1	7438	100.6 ± 1.0	15.3 ± 0.02
I1	7433	103.3 ± 1.0	15.7 ± 0.02

from the top electrode allowing a W probe to make contact through the oxide and electrically contact the bottom electrode.

An Andeen–Hagerling 2550A ultra-precision capacitance bridge is used to measure the capacitance between the top electrode and the bottom-gate electrode. The measurement is carried out at room temperature in a probe station. The stray capacitance of the probe station and experimental wiring is measured and is found to be insignificant when compared to the values obtained from the microfabricated capacitors. The excitation voltage applied by the bridge is limited to 1 V RMS to reduce the risk of dielectric breakdown. ϵ_r is thus determined by:

$$\epsilon_r = \frac{Cd}{\epsilon_0 A}, \quad (7.1)$$

where $d = 10$ nm is the oxide thickness, ϵ_0 is the permittivity of free space, and A is overlapping area of the top and bottom electrode measured using microscopy. The obtained values are shown in Table 1. An average value of $\epsilon_r = 15.41$ is used in all length calculations.

Since the distance between the ribbon and the gate is comparable to the ribbon's width we would need to employ the general formula for retrieving the length from the charging energy [216], i.e.

$$l = \frac{e^2 \left(8d \arctan \frac{w}{4d} + w \log 1 + \frac{16d^2}{w^2} \right)}{4\pi E_C \epsilon_r \epsilon_0} \quad (7.2)$$

By using this formula, we estimate lengths for the sample we have measured of about 25 nm for $E_C \simeq 40$ meV.

This section presents several devices with FET characteristics. All measurements were taken in ambient conditions with a semi-automated, 3-probe system Cascade Microtech Summit 12000 as described in Ch. 3. CuBe probes with tip diameters of $30\text{ }\mu\text{m}$ were used to contact the source and drain Au pads, while W probes with tip diameters of $10\text{ }\mu\text{m}$ were used for the gate probe. Figure 8 shows an example of a device we obtained.

The device exhibits p-type doping and the on-state channel conductance G normalised by the width of the GNR ($w_{\text{GNR}} = 1.2\text{ nm}$) increases from 0.05 S mm^{-1} at $V_{\text{SD}} = 0.1\text{ V}$ to 0.1 S mm^{-1} when V_{SD} is increased to 0.5 V ; the on-state conductivity $\sigma = GL/w_{\text{GNR}} = 3.73\text{ }\mu\text{S}$ for a channel length $L \simeq 40\text{ nm}$ (the length of the channel is estimated by averaging the transport data of multiple devices at low temperature).

The device was switched off when the gate voltage V_{G} was larger than 0.5 V and had good $I_{\text{on/off}} \simeq 10^3$ as the source-drain voltage was set to 0.1 V . Off-state conductance was estimated around $G_{\text{off}} \simeq 6\text{ pS nm}^{-1}$ at $V_{\text{SD}} = 0.1\text{ V}$ and increasing as the V_{SD} was increased, despite exhibiting consistent switching behaviour of about $SS \simeq 400\text{ mV dec}^{-1}$ and indicating good gate control over the conduction channel.

On top of that, almost no hysteresis was measured under ambient conditions. The hydrophobic nature of the sandwich established by the ribbon and graphene leads considerably minimises the undesired hysteresis in the transfer characteristics of the transistors, which is caused by the presence of water close to the GNR-dielectric contact (gate-voltage sweep rate was 44 mV s^{-1}), and outperforming early carbon nanotube FET [58].

While the on-state current is not directly related to the bandgap energy E_g , the off-state current strongly depends on E_g [22]. Assuming the synthesised pGNR is an intrinsic semiconductor and its band structure has a mid-gap alignment with the source-drain contacts, the smallest (off-state) current will occur when both the conduction and valence bands are flat. If transport is controlled by thermal carrier emission, the on-off ratio will depend exponentially on the temperature. It is then possible to give an estimate of E_g for a devices, which in case of device c1 is

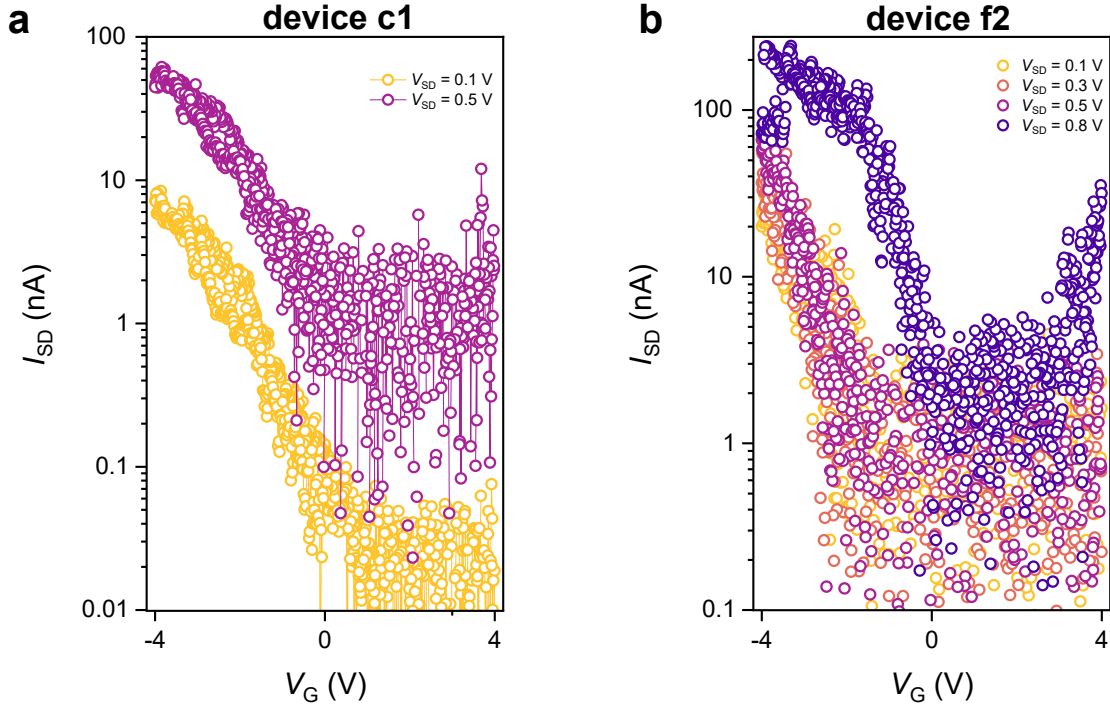


Figure 7.7: **a**, Semi-log plot of the transfer curves for a p-type FET device obtained with pGNR acquired at low and medium biases. **b**, Another p-type device with the transfer curves acquired under equispaced V_{SD} voltages.

$$2k_B T \log \left\{ \frac{I_{on}}{I_{off}} \right\} = E_g \simeq 327 \text{ meV}, \quad (7.3)$$

Since it is typically difficult to establish the threshold voltage in transfer characteristics without ambiguity, conventional FETs are frequently compared using the so-called field-effect mobility. This is device-specific rather than material-specific, and includes effects such as contact resistances and surface effects [252]. We have estimated both the linear and the saturation FET mobilities for several devices.

The conduction channel is pinched off when $V_{SD} = V_G - V_{Th}$, where V_{Th} is the threshold voltage at which the device switches on. In this condition, the current cannot increase substantially anymore and saturates to a value $I_{SD,sat}$ identifiable as a plateau by either plotting the semi-logarithmic plot of transfer characteristics or by simply visualising the output characteristics. If channel-shortening effects caused by the depletion region at the drain are neglected, the saturation current

can be obtained by substituting V_{SD} with $V_G - V_{Th}$, and within the gradual-channel approximation the drain-saturation current may be written as [253]

$$I_{SD,sat} = \frac{w_{GNR}}{2L} \mu_{sat} C_G (V_G - V_{Th})^2, \quad (7.4)$$

where C_G is the gate capacitance per unit area. From the capacitance measurements detailed above we extract a $C_G = 1.4 \times 10^{-2} \text{ F m}^{-2}$. We estimate a saturation mobility (also called field-effect mobility) for this particular device to be $0.8 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. We further estimate the field-effect mobility in the linear regime ($V_G \gg V_{SD}$) by extracting the gradient of I_{SD} versus V_G at constant $V_{SD} = 0.1 \text{ V}$

$$\mu_{lin} = \frac{\partial I_{SD}}{\partial V_G} \frac{L}{w_{GNR} C_G V_{SD}} = \frac{g_m L^2}{C_{GS} V_{SD}}. \quad (7.5)$$

We remark that the capacitance values C_G and C_{GS} : the former represents the capacitance between the nanoribbon segment that spans across the gate and the gate itself. Therefore, we have estimated that the capacitance per unit area for a ribbon of $L \simeq 40 \text{ nm}$ and $w_{GNR} = 1.2 \text{ nm}$ placed on top of a gate of thickness $d \simeq 10 \text{ nm}$ is

$$C_{GS} = \frac{\varepsilon_0 \varepsilon_r A}{d} = 0.7 \text{ aF}, \quad (7.6)$$

For device c1 we estimate a linear FET mobility $\mu_{lin} \simeq 4 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

Device f2 in Fig. 7.7 reproduces qualitatively a similar behaviour. In comparison with device c1, this device exhibits $I_{on}/I_{off} \simeq 10^2$, which then would correspond to a $E_g \simeq 220 \text{ meV}$ by using Eq. 7.3. The peculiarity of this device is that a current saturation limit is reached only at $V_{SD} = 0.8 \text{ V}$, with a $\Delta V_{gap} \simeq 3 \text{ V}$, and the off-state current tends to increase with increasing bias. The gate has good control ($SS \simeq 400 \text{ mV dec}^{-1}$) over the conductive channel at small biases, and excellent control $SS \simeq 160 \text{ mV dec}^{-1}$ at $V_{SD} = 0.8 \text{ V}$, which is amongst the highest values measured across our devices. Furthermore, the on-state channel conductance G normalised by the channel width reaches values as high as 0.2 S mm^{-1} and on-state conductivity $\sigma \simeq 6 \mu\text{S}$. We report slightly higher values for the mobility of this device as we extracted $\mu_{sat} = 3.5 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_{lin} = 16 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

Table 7.4: Linear and saturation mobilities extracted from the transfer characteristics of several devices.

Device name	μ_{lin} ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	μ_{sat} ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)
E5	5	1
C1	4	0.8
F2	16	3.5
C6	39	4
C4	11	4.3

In Table 3, we present the linear and saturation mobilities for a variety of devices. Since our pGNR has molecularly defined edges, we explain these values by assuming that the effective mobility decreases as the ribbon width approaches sub-5 nm. The linear-mobility values correlate well with those expected by theoretical estimation [85], but the saturation-mobility values have not yet been found. At room temperature, the effective mobilities are predicted to be limited by acoustic phonons. Nonetheless, if the surface-impurity density is large, this may become the major scattering mechanism, and values in the range of $10\text{--}1000 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$ are expected [254], which is consistent with the values we measured.

7.5 Concluding Remarks

7.6 Project Overview

PGNRs were provided by Dr Qiang Chen and prof. Harry Anderson. Fanmiao Kong assisted me in configuring the electronic computations on the ARC supercomputing centre, which we thank for its cooperation and assistance in troubleshooting issues when jobs were not running.

8

Conclusions and Perspectives

In this thesis, I have investigated the electronic transport properties at the device level of molecular graphene nanoribbons, focusing on those that are relevant to their potential use as a quantum technology platform and as channel material in miniaturised all-carbon logic components. In particular, I sought to verify the existence of electronic states robust enough to overcome the thermal broadening and to address the possibility to tune the electronic properties by adding chemical groups either along the edges or within the honeycomb lattice.

We established in Ch. 5 that the temperature dependence of graphene nanoribbon conductance follows a power-law, and that the molecular precision with which MGNRs can currently be synthesised allows quantum effects such as tunnelling events to be detected at temperatures close to room temperature. We attributed this behaviour to the length/width ratio of the ribbon strands, as well as the fact that long-range finite Coulomb interactions appear to play a role when measuring electronic properties, which was previously unreachable with STM experiments. A further step would be to determine whether the observed behaviour for MGNRs is due to phase transitions that they undergo to or to the presence of defects inside the MGNR lattice. To accomplish this, a longer ribbon would need to be studied, as well as possibly adding another gate at the bottom to see if it is possible to create a double

dot in a molecular system for the first time, which would represent a significant step forward in the creation of molecular qubits operating at the device level.

We demonstrated in Ch. 6 that the insertion of 3D bulky groups along the edges could be a helpful method for affording MGNR-based electrical devices where electronic and vibrational states can be examined, thus increasing the reproducibility of these types of devices. We were able to observe a strong suppression of the tunnelling current at low bias in combination with vibron-assisted excitations using milliKelvin transport spectroscopy, which was explained within the Franck–Condon model supported by DFT studies about the changes induced in the electronic and phonon dispersions by the maleimide groups. Our findings indicate that the vibrational modes in GNR are more tightly related to the electric charge than was previously reported for a class of devices built using suspended carbon nanotubes. Our statistical examination of the role of the bulky groups on a variety of devices was inconclusive due to the small sample size and the limited number of devices that could be cooled at the same time. It would thus be necessary to perform a more thorough statistical analysis using a cryogenic multiplexer, which would allow us to increase the number of measurable devices with a single cooldown. From a basic and technological standpoint, the discoveries gained in this chapter pave the way for the employment in spintronic devices not only of localised magnetic centres, as in single molecule magnets, but also of magnetic states of delocalised electronic systems, such as graphene nanoribbons.

In Ch. 7.1, we looked at two distinct methods for doping graphene nanoribbons. We predict that if large changes in the Fermi energy position of the graphene nanoribbon energy level are desired, chemical groups acting on the π -electron system should be used, and that electron donating groups induce relative position adjustments that will be extremely difficult to detect at the device level. However, the incorporation of porphyrin complexes within the graphene honeycomb appears to be a more valuable choice for coupling the spin and charge degrees of freedom without losing the electrical properties. We discover that the intrinsic mobilities caused by the addition of Ni-based porphyrin rings are comparable to those reported

for pure MGNR, that the switching behaviour for certain devices is low enough to use these hybrids as channel material in FET devices, and that the off-state currents are already competitive with those of most CNT-based FETs.

Finally, we note three issues and open concerns that emerged from this research and should be addressed in future ones. All the graphene nanoribbons used in this thesis were synthesised with methods producing ribbons of different lengths. This can be issue because the reproducibility for both SETs and FETs depends on the lateral extension of the MGNRs. Moreover, longer ribbon would be required for the construction of microprocessors, as as the one recently created entirely from carbon nanotubes [255]. Studies exploring ribbons from a particular separation chromatography could elucidate and confirm the presence or absence of finite-range interaction and whether there is a cutoff length to be fulfilled in order to verify if length is a critical parameter. Second, while the formation of GNR–GNR heterojunctions has demonstrated the formation of topologically non-trivial phases [256], it would be interesting to see if the introduction of chemical groups along the edges or inside the honeycomb lattice will induce a particle–hole symmetry breaking, and in the formation of new topologically non-trivial phases. Finally, the ability to perform quantum operations on single molecular qubits at the device level remains elusive. Graphene nanoribbons could be employed in a single-molecule EPR transport experiment, where microwave pulses excite and control the electron-spin coherence, and whether topologically-protected electronic states neare the Fermi energy could be used to improve their resistance agains main source or dechorence [99]. Furthermore, it would be interesting to insert graphene nanoribbons between two superconducting leads to see if the absence of backscattering inherent in graphene-derived nanostructures could help with Cooper-pair transmission.

Appendices



Tb–Dy Molecular Spin Valve

Contents

A.1 Introduction	157
A.2 Experimental and Computational Methods	159
A.3 Results	160
A.3.1 Computational	160
A.3.2 Coulomb-Blockade Spectroscopy	162
A.4 Concluding Remarks	163
A.5 Project Overview	165

A.1 Introduction

Typically a spin valve is a trilayered device in which a paramagnetic material (also called *spacer*) of interest is sandwiched between two ferromagnetic (FM) electrodes of different coercitivities, so they may be independetly controlled by just one magnetic field.

One of the FM acts as a *spin injector*, meaning that, if an electrical bias is applied, spins from a population P_0 are injected from the quasi-Fermi level into the paramagnet. The second paramagnet is called *spin detector* and owns its name to the fact that it provides unequal spin-up and spin-down DOS at the Fermi level, thus preferentially transmitting spins of one particular orientation. Overall this